

# Group Management in Linux — Complete Detailed Guide

Linux is a **multi-user operating system**, and groups are used to:

- Manage permissions easily
- Provide shared access to files
- Control access to applications
- Simplify administration

Instead of giving permissions to individual users one by one, we assign permissions to a **group**, and then add users to that group.

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## What is a Group?

A **group** is a collection of users.

Example:

- Developers → **devops**
- Database team → **dba**
- HR team → **hr**

If a file belongs to a group:

- All users in that group can access it (depending on permissions).
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## 2 Types of Groups in Linux

### ◆ 1. Primary Group

- Every user has exactly **one primary group**
- When a user creates a file, it belongs to their primary group

Check:

```
id username
```

Example:

```
uid=1001(ashvi) gid=1001(ashvi)
```

Here:

- Primary group = ashvi
- 

### ◆ 2. Secondary (Supplementary) Groups

- A user can belong to multiple secondary groups
- Used to provide additional access

Example:

```
uid=1001(ashvi) gid=1001(ashvi) groups=1001(ashvi),1002(devops)
```

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## 3 Important Group File

## **/etc/group**

Stores group information.

View:

```
cat /etc/group
```

Example entry:

```
devops:x:1002:ashvi,john
```

Format:

```
group_name : password_placeholder : GID : members
```

Fields:

- group\_name → Name of group
  - x → Password placeholder
  - GID → Group ID
  - members → Users in that group
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## **4** Creating Groups

### ◆ Create a Group

```
sudo groupadd devops
```

### ◆ Create a System Group

```
sudo groupadd -r devops
```

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### ◆ Create Group with Specific GID

```
sudo groupadd -g 1050 devops
```

Check group:

```
cat /etc/group
```

OR

```
getent group devops
```

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## 5 Adding Users to Groups

### ◆ Add User to Secondary Group

```
sudo usermod -aG devops ashvi
```

⚠ Important:

- `-a` → Append (do not remove existing groups)
- `-G` → Secondary group

Verify:

```
id ashvi
```

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### ◆ Set Primary Group

```
sudo usermod -g devops ashvi
```

Now:

- devops becomes primary group.
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## **6 Manage Group Members**

This command is very important for adding/removing users from groups.

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### ◆ **Add User to Group**

```
sudo gpasswd -a username groupname
```

Example:

```
sudo gpasswd -a ashvini devops
```

Now check:

```
grep devops /etc/group
```

Output:

```
devops:x:1005:ashvini
```

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### ◆ **Remove User from Group**

```
sudo gpasswd -d username groupname
```

Example:

```
sudo gpasswd -d ashvini devops
```

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### ◆ **Set Group Password (Rarely Used)**

**Set group password:**

```
sudo gpasswd devops
```

**Remove group password:**

```
sudo gpasswd -r devops
```

**This stores a password for that group in: /etc/gshadow**

```
group:password:group_admins:members
```

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**This allows users to temporarily join group using:**

```
newgrp groupname
```

**But group passwords are rarely used in modern systems.**

## When Was This Used?

This feature was useful in older Unix systems where:

- Admin didn't want to manually add users to groups
- But wanted controlled shared access
- Multiple teams shared machines

Example (old-school setup):

- Developers share a server
  - Temporary access to release group
  - Share group password internally
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## Why It's Rarely Used Today

Because it has problems:

- 1 Sharing group passwords is insecure
- 2 No proper auditing

③ Hard to track who knows the password

④ Modern systems prefer:

- `usermod -aG`
- ACLs
- sudo rules

Most companies disable this practice completely.

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## ◆ Set Group Administrators

`sudo gpasswd -A username groupname`

Group admin can manage members without root.

Example:

`sudo gpasswd -A user1 devops`

A **group admin** is a user who can manage the group **without being root**.

They can:

- Add users to the group
- Remove users from the group
- Set/remove group password (if allowed)

They CANNOT:

- Delete the group
- Rename the group
- Change GID  
(Those require root privileges)

## Practical Example

### Step 1: Create group

```
sudo groupadd devops
```

### Step 2: Make user1 group admin

```
sudo gpasswd -A user1 devops
```

### Step 3: Switch to user1

```
su - user1
```

Now user1 can manage group members:

```
gpasswd -a user2 devops
```

```
gpasswd -d user2 devops
```

Without sudo.

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## Why Is This Rarely Used?

In modern systems:

- Admin tasks are done via sudo
- Not via group administrators

So `gpasswd -A` is mostly legacy functionality.

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## Important Difference

| Option | Purpose |
|--------|---------|
|--------|---------|

|    |                          |
|----|--------------------------|
| -A | Set group administrators |
| -a | Add user to group        |
| -d | Delete user from group   |
| -r | Remove group password    |

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## **7** Viewing Group Membership

**Check group members:**

```
getent group devops
```

**Check user's groups:**

```
groups ashvi
```

OR

```
id ashvi
```

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## **8** Modifying Group Details

**groupmod** = modify an existing group

It is used to:

- Rename a group
- Change group GID
- Change group password (rarely used)

## Syntax

```
sudo groupmod [options] groupname
```

### ◆ Most Used Options

#### 1 -n → Rename a group

```
sudo groupmod -n newname oldname
```

#### Example

```
sudo groupmod -n devteam devops
```

Now:

```
getent group devteam
```

#### ⚠ Important:

- GID stays the same
  - Users remain in the group
  - Only name changes
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#### 2 -g → Change GID (Group ID)

```
sudo groupmod -g NEW_GID groupname
```

#### Example

```
sudo groupmod -g 2005 devteam
```

Now:

```
getent group devteam
```

You will see new GID.

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### Important About Changing GID

If files were owned by old GID:

```
ls -l file.txt
```

They may show numeric group ID instead of group name.

To fix file ownership after GID change:

```
sudo find / -group OLD_GID -exec chgrp devteam {} \;
```

**In real production, changing GID should be done carefully.**

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## -p → Change Group Password

```
sudo groupmod -p encrypted_password groupname
```

 Rarely used directly because:

- Password must be encrypted
  - Usually we use `gpsswd` instead
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## 9 Deleting Groups

### Delete group

```
sudo groupdel devops
```

 Cannot delete if it is primary group of any user.

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## 10 Default Group Settings

When a new user is created:

- A private group is created (same name as user)
- This is defined by settings in:

 **/etc/login.defs**

Check:

```
cat /etc/login.defs
```

Look for:

```
USERGROUPS_ENAB yes
```

If enabled:

- Each new user gets a private group.

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## 11 Important Commands Summary

| Command        | Purpose              |
|----------------|----------------------|
| groupadd       | Create group         |
| groupdel       | Delete group         |
| usermod<br>-aG | Add user to group    |
| usermod -g     | Change primary group |

`gpasswd -d`    Remove user from  
group

`groups`        Show user groups

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## Common Mistakes

 Using:

```
usermod -G devops ashvi
```

Without `-a`

This removes user from all other groups.

Correct:

```
usermod -aG devops ashvi
```

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## Best Practices

- ✓ Use groups instead of giving permissions to individuals
  - ✓ Use SGID for shared directories
  - ✓ Regularly audit group memberships
  - ✓ Avoid unnecessary group privileges
  - ✓ Use meaningful group names
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